BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Two-Dimensional Arrays

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. How can a loop total an array's values?

2. What is an embedded system designed for?

A) Any desktop application
B) Only browsing websites
C) A specific task
D) Only storing backups

3. "Adding active users always increases the bandwidth available to each user." Correct this. Explain why.

4. A school needs central accounts and shared files. Three club laptops share files directly. Correct both claims. Explain why. (a) Every client must be a dedicated server. (b) Peer-to-peer needs a central server.

Today's objectives

Store and access two-dimensional array values. Use the technique in an independent scenario.

Key vocabulary - Read, cover, recall

2D array
Stores data in a fixed-size grid accessed by row and column indices.

Row index
The first index identifies a particular row; indexing starts at zero.

Column index
The second index identifies the column within the chosen row.

Grid assignment
A value is set or retrieved using both its row and column coordinates.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

2D array	Stores data in a fixed-size grid accessed by row and column indices. Apply this in a theatre seats.
Row index	The first index identifies a particular row; indexing starts at zero. Apply this in a school seating plan.
Column index	The second index identifies the column within the chosen row. Apply this in a game board.
Grid assignment	A value is set or retrieved using both its row and column coordinates. Apply this in a class timetable.

I Do - One worked example

A theatre seats is being developed. Explain how 2d array and row index are used. [4 marks]

1. Read the scenario and identify the data the program needs.

2. 2D array [1]; Stores data in a fixed-size grid accessed by row and column indices. [1].

3. Row index [1]; The first index identifies a particular row; indexing starts at zero. [1].

4. Two linked points for each concept.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A school seating plan uses 2d array. Explain how it helps. [2 marks]

2. Explain row index in a game board. [2 marks]

3. A class timetable uses column index and grid assignment. Explain both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Print the label at row 1, column 0.

```
array desk[2,2]
desk[0,0] = "PC1"
desk[1,0] = "PC2"
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State TWO relevant techniques from this topic. [2 marks]

Q3. Explain how row index and column index help with a museum room matrix. [4 marks]

Q4. A programmer designs a parking spaces. Explain why 2d array, column index and grid assignment matter. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A two-dimensional array stores data in a grid accessed by row and column indexes. ☐ T ☐ F
2. Stores data in a fixed-size grid accessed by row and column indices ☐ T ☐ F
3. The first index identifies a particular row; indexing starts at zero ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Set a total to 0. Loop through the array, adding each value once. ☐

Q2 C - A specific task ☐

Q3 Users may compete for finite bandwidth, reducing the data rate available to each. ☐

Q4 (a) A server manages shared services and resources for clients. (b) Peers share directly and can act as clients and servers. Central control suits a school; peer-to-peer can cost less for a small club but is harder to manage as it grows. ☐

Code First - Model

```
array desk[2,2]
desk[0,0] = "PC1"
desk[1,0] = "PC2"
print(desk[1,0])
```

Output: PC2

We Do - Check the model

1. 2D array [1]; Stores data in a fixed-size grid accessed by row and column indices. [1]. ☐

2. Row index [1]; The first index identifies a particular row; indexing starts at zero. [1]. ☐

3. Column index [1]; The second index identifies the column within the chosen row. [1]. Grid assignment [1]; A value is set or retrieved using both its row and column coordinates. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 2D array [1]; Column index [1]. ☐

Q3 Row index [1]; The first index identifies a particular row; indexing starts at zero. [1]. Column index [1]; The second index identifies the column within the chosen row. [1]. ☐

Q4 2D array [1]; Stores data in a fixed-size grid accessed by row and column indices. [1]. Column index [1]; The second index identifies the column within the chosen row. [1]. Grid assignment [1]; A value is set or retrieved using both its row and column coordinates. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: